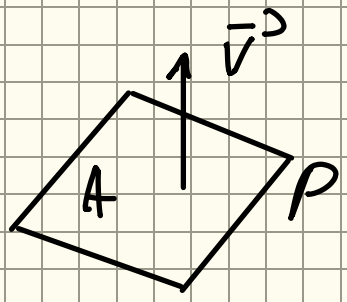
1) Write the equation for the plane through $A(2,-3,0)$ and it is $\perp$ $\vec{v} = \vec{i} - \vec{j} + \vec{k}$

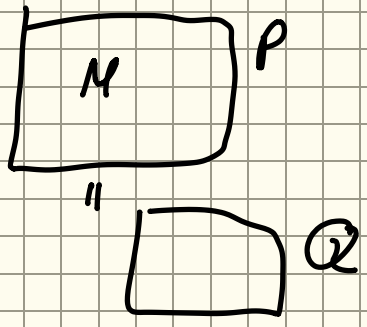
$A(x_A, y_A, z_A)$ |

$\vec{n} \perp P$




$\Rightarrow P: a(x-x_0)+b(y-y_0)+c(z-z_0)=0$

2) Determine the equation of the plane through the point $M(1,-1,2)$ and which is $\parallel$ to $Q: 2x+y-4z+5=0$



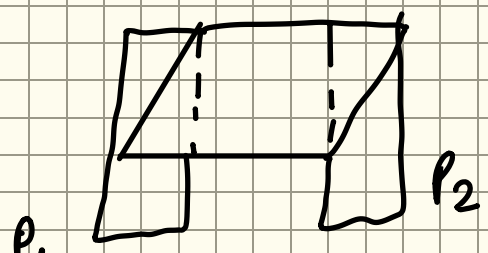

$\Rightarrow \vec{v}_Q = 2\vec{i} + \vec{j} - 4\vec{k}$

$2(x-1)+y+1-4(z-2)=0$

$2x-2+y+1-4z+8=0$

$2x+y-4z+7=0$

4) Write the equation of the plane $P$ through $M(2,-3,1)$ $\perp$ to both $P_1: x-y+2z-5=0$ and $P_2: 2x-3y-z+4=0$




$\vec{v}_1 = \vec{i} - \vec{j} + 2\vec{k}$

$\vec{v}_2 = 2\vec{i} - 3\vec{j} - \vec{k}$

$P \perp P_1 \mid \vec{v}_1 \perp P_1 \Rightarrow \vec{v}_1 \parallel P$

$P \parallel P_2 \mid \vec{v}_2 \perp P_2 \Rightarrow \vec{v}_2 \parallel P$

$M \in P$

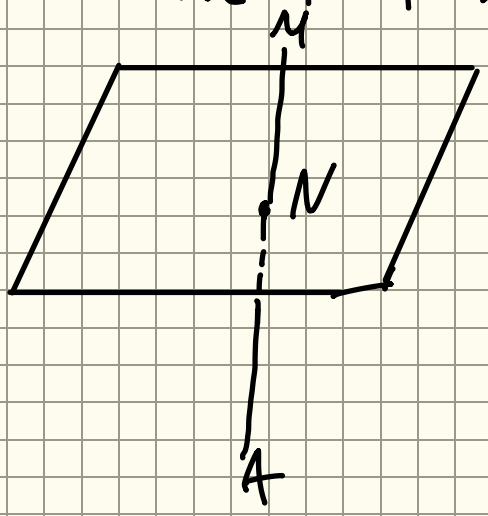
$\Rightarrow$

$$P:\begin{vmatrix} x-2 & y+3 & z-1 \\ 1 & -1 & 2 \\ 2 & -3 & -1 \end{vmatrix} = (x-2)\cdot 7 + 5(y+3) - z + 1 = 0$$

$$\Leftrightarrow 7x - 14 + 5y + 15 - z + 1 = 0$$

$$\Leftrightarrow 7x + 5y - z + 2 = 0$$

5) Determine the equation of the plane with respect to which $M(2, -3, 5)$ and $A(-6, 1, 3)$ are symmetric




$N(-2, -1, 4)$

$\vec{AM} \perp P \Rightarrow \vec{AM} = 8\vec{i} - 4\vec{j} + 2\vec{k}$

$\Rightarrow P: 8(x+2) - 4(y+1) + 2(z-4) = 0$

$8x + 16 - 4y - 4 + 2z - 8 = 0$

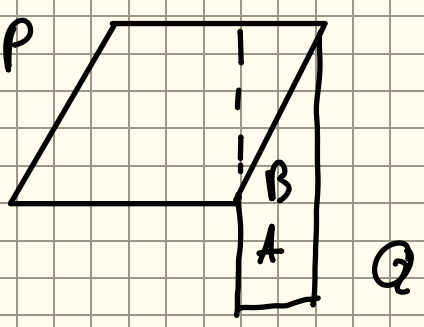
$8x + 4 - 4y + 2z = 0$

$4x - 2y + z + 2 = 0$

6) Write the equation of a plane through $A(2, -1, 3)$ and $B(1, -3, 5)$ in each of the following alternative case:

1. The plane in $\perp$ to $P: x - y + 3z + 5 = 0$
2. The plane in $\parallel$ to the vector $\vec{v} = 2\vec{i} - 3\vec{j} + 4\vec{k}$

1.




$\vec{n} = \vec{i} - \vec{j} + 3\vec{k}$

$\left.\begin{matrix} \vec{n} \perp P \\ Q \perp P \end{matrix}\right| \Rightarrow \vec{n} \parallel Q$

$A, B \in Q \Rightarrow \vec{AB} \parallel Q$

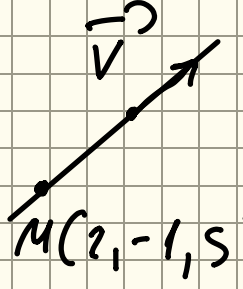
$\vec{AB} = -\vec{i} - 2\vec{j} + 2\vec{k}$

$$Q: \begin{vmatrix} x-2 & y+1 & z-3 \\ 1 & -1 & 3 \\ -1 & -2 & 2 \end{vmatrix} \Leftrightarrow 4(x-2) - 5(y+1) - (z-3) = 0$$
$$\Leftrightarrow 4x - 8 - 5y - 5 - z + 3 = 0$$
$$4x - 5y - z - 10 = 0$$

2. $\vec{AB} = -\vec{i} - 2\vec{j} + 2\vec{k}$

$$Q: \begin{vmatrix} x-2 & y+1 & z-3 \\ 2 & -3 & 4 \\ -1 & -2 & 2 \end{vmatrix} \Leftrightarrow 2(x-2) - 8(y+1) - (z-3) = 0$$
$$\Leftrightarrow 2x - 4 - 8y - 8 - z + 3 = 0$$
$$2x - 8y - z - 9 = 0$$

7) Write the equation of the line through the point $M(2,-1,5)$ and $\parallel$ to $\vec{v} = 3\vec{i} - 5\vec{j} + \vec{k}$



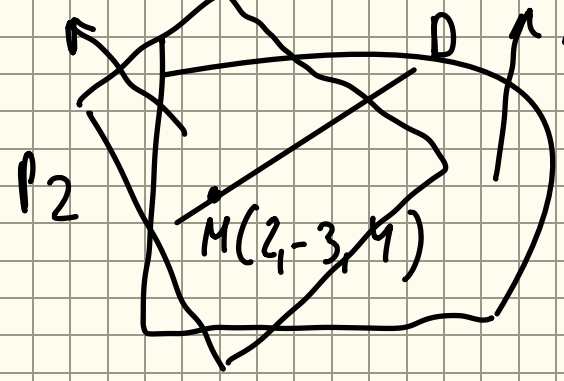

$$M(x_M, y_M, z_M) \Rightarrow \begin{cases} x = x_M + \ell t \\ y = y_M + m t \\ z = z_M + n t \end{cases}$$
$$\vec{v}(\ell, m, n)$$

$$\Leftrightarrow \frac{x - x_M}{\ell} = \frac{y - y_M}{m} = \frac{z - z_M}{n} = t$$

$$D: \frac{x-2}{3} = \frac{y+1}{-5} = \frac{z-5}{1}$$

8) Write the equation of the line through $M(2,-3,4)$ parallel to $d: \begin{cases} x - y + z - 3 = 0 & (P_1) \\ x + y - 4 = 0 & (P_2) \end{cases}$

$\vec{n_2}(1,1,0)$




$\vec{n_1} \perp P_1$, $D \subset P_1$ $\Rightarrow \vec{n_1} \perp D$

$\vec{n_2} \perp P_2$, $D \subset P_2$ $\Rightarrow \vec{n_2} \perp D$

$\Rightarrow D \perp$ plane with $\vec{n_1}, \vec{n_2}$

$\vec{n_1} \times \vec{n_2} \perp$ plane $\vec{n_1}, \vec{n_2}$ $\Rightarrow D \parallel \vec{n_1} \times \vec{n_2}$

$$\vec{n_1} \times \vec{n_2} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 1 \\ 1 & 1 & 0 \end{vmatrix} = -\vec{i} + \vec{j} + 2\vec{k}$$

$$D: \frac{x-2}{-1} = \frac{y+3}{1} = \frac{z-4}{2} = t$$

g) Write the equation of the segment $[AB]$ with $A(2,-1,3)$ and $B(-2,1,5)$

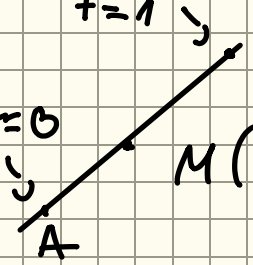
$$\frac{x-x_A}{x_B-x_A} = \frac{y-y_A}{y_B-y_A} = \frac{z-z_A}{z_B-z_A}$$

$$\frac{x-2}{-4} = \frac{y+1}{2} = \frac{z-3}{2} = t \quad (1)$$

$\Rightarrow x = 2-4t$

(1) $y = -1+2t$ ($t=0$ at $A$, $t=1$ at $B$)

$z = 3+2t$

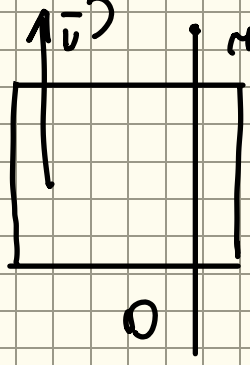



$\Rightarrow t=0 \Rightarrow \begin{cases} x_0 = 2 \\ y_0 = -1 \\ z_0 = 3 \end{cases} = A$

$t=1 \Rightarrow \begin{cases} x=-2 \\ y=-1 \\ z=5 \end{cases} = B$

$\Rightarrow [AB] = (1) \cap (2),\ t \in [0,1]$

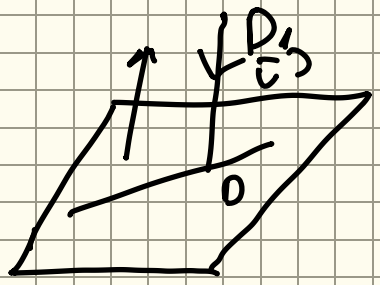
10) Write the equations of the line to the point $M(2,1,-3)$ which is $\perp P: x-3y-7z+5=0$




$P$ $\quad \vec{n} \perp P,\ D \perp P \Rightarrow D \parallel \vec{n},\ M \in D \Rightarrow$

$$\frac{x-2}{1} = \frac{y-1}{-3} = \frac{z+3}{-7} = t$$

11) Write the equation of a line contained in $P: x-y+3z-5=0$ and $\perp$ to $D_1: \begin{cases} x = 3+t \\ y = 2-t \\ z = 5 \end{cases}, t \in \mathbb{R}$ and $M(1,-1,1)$




$\vec{n}(1,-1,3)$ $\quad \vec{n} \perp P,\ D \subset P \Rightarrow \vec{n} \parallel D$

$\vec{v}(1,-1,0)$ $\quad \vec{v} \subset \alpha_1,\ \alpha_1 \perp P \Rightarrow \vec{v} \perp D$

$\Rightarrow D \perp$ plane $\vec{v}\,\vec{n}$

$\vec{n} \times \vec{v} \perp$ plane $\vec{v}\,\vec{n} \Rightarrow D \parallel \vec{n} \times \vec{v}$

$$\vec{n} \times \vec{v} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & -1 & 3 \\ 1 & -1 & 0 \end{vmatrix} = 3\vec{i} + 3\vec{j}$$

$$\Rightarrow \frac{x-1}{3} = \frac{y+1}{3} = \frac{z-1}{0} = t \Rightarrow z-1=0$$

$$\Rightarrow D: \begin{cases} \frac{x-3}{3} = \frac{y+1}{3} \end{cases}$$

$1 \; 2 = 1$